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Tailoring explanations in conversational recommendations: The impact of decision contexts and user interfaces

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ABSTRACT

Explainability is crucial for building trust in traditional recommendation systems, yet its role in conversational settings is underexplored. Across three experimental studies (N = 1,429), we used between-subjects designs featuring diverse product categories (cameras, smartwatches, headphones) to examine the interactive effects of post hoc explanations (expert validation-based vs. consensus validation-based) and decision-making domains (hedonic vs. utilitarian) on consumer responses to conversational recommendations. We further examined how consumer decision-making styles (intuitive vs. rational) and user interfaces (text-based vs. voice-based) moderated these effects. Results show that post hoc explanations enhance perceived transparency and interpretability, thereby increasing consumer trust in conversational recommendations. In text-based interfaces, consumers making hedonic decisions preferred consensus-based explanations, whereas no clear preference emerged for utilitarian decision-makers. In voice-based interfaces, utilitarian consumers favored consensus-based explanations, while no significant preference was observed for hedonic decisions. Furthermore, intuitive consumers preferred consensus-based explanations for hedonic decisions and expert-based explanations for utilitarian decisions. Rational consumers consistently favored consensus-based explanations across both decision-making domains. These findings provide valuable insights for designing conversational recommendation systems on e-commerce platforms. By tailoring explanations to decision domains, user interfaces, and consumer decision-making styles, businesses can foster greater trust and engagement, driving more favorable purchasing behaviors and improving business outcomes.

1. Introduction

Conversational recommendation agents, driven by advancements in Generative Artificial Intelligence (AI), are transforming user engagement and personalization on digital platforms. These agents provide tailored responses and highly personalized recommendations through dialogue (Pramod and Bafna, 2022). By managing complex inquiries and adapting responses to individual preferences, they enhance the user experience (Chang and Park, 2024; Jannach et al., 2021). Examples like Amazon's Rufus and Alibaba's Tmall Genie demonstrate the practical impact and potential of these systems in marketing.

Despite these benefits, conversational recommendation systems also pose challenges. A key issue is their "black box" nature, which makes it difficult for users to understand why certain recommendations are provided (Rai, 2020). These systems encourage users to ask questions like "Why is this option recommended?", but a lack of clear explanations can erode trust and satisfaction. Therefore, ensuring that users understand the reasoning behind recommendations is critical (Shin, 2021).

Unlike conventional recommendation systems, conversational recommendation agents leverage Natural Language Generation (NLG) and advanced dialogue management systems to provide real-time, natural language interactions. This immediacy highlights the importance of explainability. While research on explainability in static recommendation systems has expanded, studies on explainable conversational recommendations remain limited (Guo et al., 2023a,b). As generative AI adoption accelerates, understanding how consumers respond to explainability in conversational agents becomes increasingly important (Gao et al., 2021).

Explainability in recommendation systems goes beyond the explanation algorithms themselves; the design of the user interface also plays a critical role in helping users comprehend and trust recommendations (Kim et al., 2023a,b; Mucha et al., 2021). However, much of the current

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research examines explanation methods and user interfaces separately, overlooking an integrated approach that considers both aspects together (Dominguez et al., 2019). For an optimal user experience, it is essential to align explanation methods with user interface design. This fragmented research approach limits our understanding of explainability by neglecting the combined effects of explanation methods and user interfaces on consumer responses. Notably, most research on explainability in conversational recommendation has focused on text-based systems (Gao et al., 2021), while studies on voice-based explainability remain scarce (Rhee and Choi, 2020). As conversational agents gain popularity, voice-based explainability research is expected to grow.

Since conversational agents interact directly with end-users, the effectiveness of their explanations depends on tailoring them to meet users' specific needs and contexts (Miller, 2019). Recent studies on individual differences in recommendation systems emphasize the importance of considering diverse user preferences and requirements (Chinchanachokchai et al., 2021; El Aouifi et al., 2023). To ensure that recommendations are both personalized and relevant, conversational recommendations must consider consumer traits and application contexts. Decision domains, which reflect variations in consumers' motivations and informational needs, also play a key role in this process (Chen et al., 2024). Incorporating an understanding of these domains into research on recommendation system explainability is critical to providing a more personalized recommendation experience.

This research examines the impact of conversational agents on online product recommendations, focusing on key questions: Do post hoc explanations enhance consumer responses to conversational recommendations? What mechanisms drive this effect? How do post hoc explanations and decision-making domains influence consumer reactions? How do user interfaces and decision-making styles moderate these effects?

This study contributes to the literature on user-centered, explainable recommendations (see Web Appendix A). While previous research has explored the credibility, authenticity, ethical responsibility, and persuasive power of conversational agents, few studies have examined how contextual factors and individual characteristics shape consumer responses to explainability. By analyzing consumer perceptions of different explanatory approaches in conversational recommendation agents across various decision-making contexts and user interfaces, this research provides valuable insights into the mechanisms underlying their effectiveness.

2. Theoretical framework and hypotheses development

2.1. Explainable recommendation

Recommendation systems are information filtering tools designed to assist users by identifying and suggesting products or services that match their interests and preferences (Da'u & Salim, 2020). These systems are widely used in applications like online shopping, social media, and entertainment platforms (Guo et al., 2023a,b). By personalizing user experiences, recommendation systems enhance engagement and drive sales growth (Hallikainen et al., 2022; Messaoudi and Loukili, 2024).

Explainable recommendation systems extend beyond providing suggestions by offering users insights into the rationale behind recommendations (Zhang and Chen, 2020). Clear explanations can mitigate concerns about AI "black boxes," reduce perceived risk, and bolster transparency (Chinchanachokchai et al., 2021). This added transparency improves the system's persuasiveness, effectiveness, and trustworthiness, increasing user engagement and satisfaction (Shin, 2020).

Despite these advantages, explainable recommendation systems have limitations. Most provide static, single-turn explanations that neither incorporate user feedback nor adapt to contextual changes. Additionally, customizing explanations to accommodate diverse user preferences and varying levels of understanding remains a challenge. To overcome these limitations, conversational recommendation systems have gained attention (Kim et al., 2023a,b). Unlike static systems, conversational recommendation agents offer smoother, more natural interactions that capture real-time user feedback and preferences. This enables them to deliver more timely and personalized experiences (Kostric et al., 2024; Wang et al., 2023a,b).

2.2. Consumer responses to conversational recommendation explainability

Understanding user-centered explainability is crucial for meeting stakeholder needs (Kaplan et al., 2024). Drawing on Social Cognitive Theory (Bandura, 2001), which posits that individuals learn by observing others and that cognitive factors such as self-efficacy and outcome expectations shape behavior, we argue that transparency and interpretability in conversational recommendation agents can enhance positive consumer responses. Clear explanations serve as social modeling, enabling users to form accurate mental representations of the recommendation process, thereby increasing their trust and willingness to follow recommendations (Schunk, 2012; Zimmerman, 2000). Recent studies have demonstrated that when users perceive the decision-making process as transparent and easy to understand, they are more likely to engage with and adopt the system's recommendations (Shin, 2020).

Given that conversational agents offer real-time suggestions, users need to quickly grasp the reasoning behind these recommendations. Therefore, the agent must not only provide timely responses but also offer immediate and comprehensible explanations (Zaidi et al., 2024). Enhanced explainability helps consumers understand how the system addresses their preferences, improving their perceptions of the agent's transparency and interpretability (Chen et al., 2024; Pecune et al., 2019).

We define *perceived transparency* as the visibility and clarity of the conversational recommendation agent's decision-making process. High transparency allows users to understand how recommendations are generated, fostering trust and satisfaction (Friedman et al., 2023; Paz-Ruza et al., 2024). *Interpretability* refers to how easily users can comprehend the agent's recommendations, enhancing their ability to evaluate and trust the system (Chen et al., 2024). Based on these insights, we propose the following hypotheses:

H1. The presence (vs. absence) of post hoc explanations enhances positive consumer responses to conversational recommendations.

H1a. The effect of post hoc explanations on consumer responses to conversational recommendations is mediated by perceptions of transparency.

H1b. The effect of post hoc explanations on consumer responses to conversational recommendations is mediated by perceptions of interpretability.

2.3. Matching post-hoc explanations with decision-making domain

The interactive and real-time nature of conversational recommendation systems (CRSs) demands careful selection of explanation types (Wang et al., 2023a,b). Unlike static systems, CRSs engage users in dynamic dialogues, requiring explanations that not only convey information but also build trust (Wilkinson et al., 2021). Jaccard and Jacoby (2019) originally identified four types of conceptual validation: consensual, expert, internal, and empirical. This study focuses on consensual and expert validation, as they align most closely with lay users' interpretations of social cues and authoritative endorsements (Naiseh, 2024).

From a social proof perspective (Cialdini and Goldstein, 2004), consensus-based explanations enhance credibility by demonstrating broad user agreement, akin to electronic word-of-mouth (eWOM) findings that show perceived group endorsement fosters trust (Duan et al.,

2008). Expert-based explanations derive their persuasive strength from the source's credibility and expertise, consistent with research on authoritative endorsements (Ohanian, 1990). Consensual validation-based explanations, which reflect collective user opinions, are particularly effective in emotionally driven contexts or socially influenced contexts. In contrast, expert validation-based explanations leverage specialized knowledge or authority to support reliable decision-making (Tong et al., 2023).

The Fit or Matching perspective suggests that persuasion is more effective when the information style matches users' goals or motivations (Higgins, 2000). Research shows that consumers in hedonic contexts prefer subjective, emotion-driven explanations, whereas those in utilitarian contexts seek objective, fact-based explanations (Chen et al., 2024). Hedonic decision-making, driven by emotions and pleasure, leads consumers to prefer subjective, socially influenced cues (Longoni and Cian, 2022). Accordingly, consensual explanations, which emphasize community validation and emotional resonance, are especially compelling in hedonic contexts (Luo et al., 2024). In contrast, utilitarian decision-making prioritizes functionality, practicality, and efficiency (Longoni and Cian, 2022). Consumers in utilitarian contexts typically engage in central route processing, making expert-based explanations—which feature data-driven insights and credible endorsements—more persuasive (Petty, 1986; Voss et al., 2003).

Construal Level Theory further clarifies why these explanation styles complement different decision-making domains. Hedonic choices involve higher-level construal focusing on desirability and experiential benefits. As a result, user consensus-based explanations, which emphasize social proof and emotional resonance, are more persuasive (Hamilton and Thompson, 2007; Trope and Liberman, 2010). In contrast, utilitarian decisions rely on lower-level construal, emphasizing feasibility and detailed considerations, making expert-based explanations, which highlight practicality and technical merits, more effective (Kirshner, 2024; Scarpi, 2021). Based on these insights, we propose the following hypotheses:

H2. The effectiveness of post hoc explanations is enhanced by the fit between the explanation type and the decision-making domain.

H2a. For hedonic decisions, user consensus-based explanations (vs. expert-based) lead to more positive consumer responses.

H2b. For utilitarian decisions, expert-based explanations (vs. consensus-based) lead to more positive consumer responses.

2.4. The moderating role of decision-making style

Intuitive decision-makers rely on fast, affect-driven judgments based on emotions or prior experiences, consistent with System 1 processing in Dual Process Theory. In contrast, rational decision-makers engage in deliberate, analytical reasoning, reflecting System 2 processing, which involves cognitive effort and logical evaluation (Phillips et al., 2016).

These decision-making styles influence how individuals interpret information, interact with product recommendations, and develop trust in the recommendation source (Mi et al., 2024; Virdi et al., 2020). When evaluating AI-generated explanations, consumers with a System 1 orientation respond more favorably to heuristics and emotional cues, whereas System 2-oriented consumers prefer detailed, evidence-based insights (Kahneman, 2011; Xiao and Benbasat, 2018).

Intuitive consumers, who rely on emotions and heuristics (Del Campo et al., 2016), are naturally drawn to explanations that align with their affective motives. In hedonic decision-making, where choices are driven by pleasure (Lu et al., 2016), intuitive consumers prefer explanations that incorporate social cues. According to Social Influence Theory, individuals seek guidance from others' opinions in emotionally driven contexts (Cialdini and Goldstein, 2004). Therefore, user consensus-based explanations provide the emotional validation that appeals to intuitive decision-makers in hedonic contexts.

In utilitarian decision-making, intuitive consumers tend to favor expert-validated explanations, as these simplify complex information, reduce cognitive load, and facilitate quick decision-making with minimal effort (Epstein et al., 1996). Given that intuitive (System 1) processors are influenced by simple cues that reduce uncertainty (Del Campo et al., 2016), this reinforces the matching effect proposed in H2. In contrast, rational (System 2) consumers engage in systematic evaluation and heuristic cues (Chaiken, 1980), potentially diminishing the impact of the matching effect. Since rational consumers rely on analytic processing, they may not exhibit the same sensitivity to explanation type. Therefore, we focus our hypothesis on intuitive consumers and propose the following:

H3. Intuitive decision-makers exhibit a stronger matching effect (H2) than rational decision-makers.

2.5. The moderating role of user interface

Voice-based interfaces introduce unique cognitive and social dynamics compared to text-based interfaces. They impose a higher cognitive load due to the demands of auditory processing, which limit the amount of information users can absorb at once (Ischen et al., 2022). As a result, consumers tend to prefer simpler explanations. Furthermore, voice interfaces create a heightened sense of social presence, making interactions feel more personal (Munnukka et al., 2022), which in turn influences perceptions of credibility and trust.

In utilitarian decision-making with voice-based interfaces, consumers may prefer user consensus explanations. Utilitarian decisions prioritize practicality and efficiency (Longoni and Cian, 2022). User consensus explanations simplify the decision-making process by offering quick insights based on collective opinions (Herlocker et al., 2000). Furthermore, the social presence in voice interactions enhances the communal aspect of consensus information (Van Zant and Berger, 2020). Statements such as "Many users recommend this product" foster trust in collective judgment, making consensus explanations more persuasive (Li et al., 2023). The auditory modality of voice interfaces accentuates social proof, making peer recommendations feel more relatable and trustworthy, thereby enhancing the effectiveness of consensus-based explanations.

Conversely, in hedonic decision-making within voice-based interfaces, the strong sense of social presence enhances the persuasiveness of expert validation. Hedonic decisions, driven by desires for pleasure and enjoyment, benefit from the CRS adopting an authoritative, human-like figure, which boosts the credibility of expert explanations (Nass, 2005). The personal nature of voice communication makes expert recommendations feel like trusted advice, increasing both enjoyment and confidence. Moreover, emotional cues in voice—such as tone and inflection—intensify the impact of expert validation, aligning with the Elaboration Likelihood Model, which posits that consumers rely on peripheral cues, such as perceived authority, in emotionally engaging contexts (Petty et al., 1983). Therefore, in voice-based hedonic contexts, expert-based explanations become more influential due to the enhanced social presence and emotional resonance of the interface. Based on these theoretical foundations, we propose the following hypotheses:

H4. The user interface (text-based vs. voice-based) moderates the matching effect.

H4a. Consumers making hedonic decisions are more likely to prefer expert - based explanations in voice-based interfaces.

H4b. Consumers making utilitarian decisions are more likely to prefer user consensus-based explanations in voice-based interfaces.

Fig. 1 presents the research model, illustrating the relationships among the key variables.

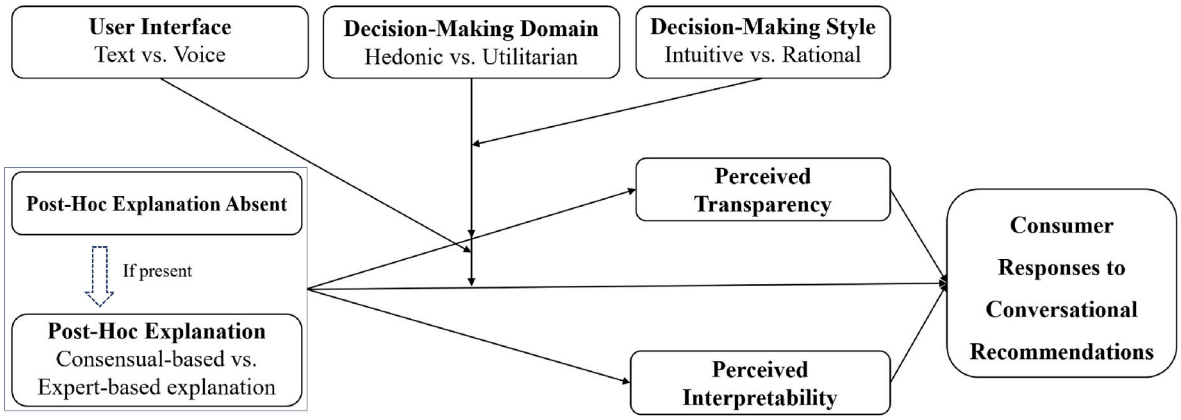


Fig. 1. Research model.

3. Overview of studies

We conducted three studies to investigate how post hoc explanations in conversational recommendations impact positive consumer responses. While each study employs a slightly different measure (e.g., trust, follow intention), all focus on assessing positive consumer responses. Study 1 tested whether the presence (vs. absence) of post hoc explanations enhances trust in the recommendation ($\mathbf{H_1}$), and whether this effect is mediated by perceived transparency ($\mathbf{H_{1a}}$) and interpretability ($\mathbf{H_{1b}}$). Study 2 examined how decision-making domains ($\mathbf{H_2}$) and decision-making styles ($\mathbf{H_3}$) moderate these effects. Study 3 investigated whether the user interface influences the relationship between explanation type and decision-making domain ($\mathbf{H_4}$) (see Table 1).

4. Sample and data collection

This research employs a quantitative methodology using an online self-administered questionnaire to investigate the hypothesized relationships. Adopting a positivist paradigm aligned with experimental research principles, we aimed to establish causal relationships by systematically controlling variables (Bryman, 2016). The target population consisted of adults aged 18 and older. To ensure diverse representation, we recruited participants through Credamo, which offers a broader and more varied sample compared to traditional recruitment methods. A random sampling approach was employed to reduce self-selection bias, thereby enhancing confidence that the final sample reflects the broader adult consumer population engaging with online platforms (see Table 2 for demographic details).

The survey introduction provided participants with an overview of the study's objectives, estimated duration, and confidentiality policies, emphasizing voluntary participation and the right to withdraw at any time without penalty. To ensure data reliability, the survey included two screening questions to confirm participant eligibility and attention checks (e.g., I have never used a mobile phone) to identify and exclude unreliable responses from the final dataset.

Table 1
Overview of studies.

Items	Study 1	Study 2	Study 3
Goal	H_1	H_2, H_3	H_2, H_4
Design	One-way between-subjects	Two-way between-subjects	Three-way between-subjects
Manipulation	Post hoc explanation (absent vs. present)	2 (explanation type: consensus vs. expert) x 2 (decision domain: hedonic vs. utilitarian)	2 (explanation type: consensus vs. expert) x 2 (decision domain: hedonic vs. utilitarian) x 2 (interface: text vs. voice)
Product	Camera	Smartwatch	Headphones
Measures	Interpretability, transparency, trust	Transparency, decision-making style, trust	Intention to follow the recommendation

Table 2
Demographics.

Construct	Item	Study 1 *(N = 300)*	Study 2 *(N = 400)*	Study 3 *(N = 729)*
Gender	Male	100	168	270
	Female	200	232	459
Age	<25	102	117	272
	25–34	139	210	333
	35–44	49	60	91
	45–54	9	9	23
	55 or order	1	4	10
Education	High school or below	8	4	13
	Associate degree or equivalent	12	19	27
	Bachelor	206	279	532
	Master or above	74	98	157

5. Study 1

Study 1 investigates whether providing post hoc explanations (vs. no explanation) enhances perceived transparency and interpretability, thereby boosting consumer trust in conversational recommendations.

5.1. Method

Design and sample. AI-driven recommendation systems are mainly integrated into digital retail, making online consumers the most relevant study population. We recruited 300 participants through Credamo (https://www.credamo.com/), a platform commonly used in online consumer behavior research (Chen et al., 2025). The demographic composition of our sample (ages 18 to 57, $M = 28.80$ years, $SD = 6.86$; 66.7% female) closely aligns with industry reports on e-commerce consumers in China (CNNIC, 2024), enhancing the generalizability of our findings to digital retail settings. Participants showed strong familiarity with AI technologies ($M = 5.04$, $SD = 1.04$), with 64% scoring in the high familiarity range (5–7), further supporting the relevance of our sample for studying AI-driven recommendation systems. Each participant received 3 CNY, and no participants were excluded from the study.

Procedure and measures. Participants were asked to imagine they were purchasing a mid-range camera on an online shopping platform, using the platform's conversational recommendation agent to assist in their selection. They were randomly assigned to one of three conditions: consensual validation-based explanation, expert validation-based

explanation, or no explanation. Then, they completed a survey measuring perceived interpretability (e.g., "I think the process of the recommendation of the product is clear," α = 0.91; Chen et al., 2024), perceived transparency (e.g., "I understood why the items were recommended to me," α = 0.94; Hernandez-Bocanegra and Ziegler, 2023), trust in the recommendation (e.g., "I trust the recommendation provided by the AI assistant," α = 0.95; Chen et al., 2024), AI familiarity (α = 0.87), product familiarity, and demographics. All continuous variables were rated on a 7-point scale (1 = strongly disagree, 7 = strongly agree). Web Appendix E provides the complete instructions and measurement scales.

5.2. Results

We tested whether explanations lead to more positive consumer responses than no explanation (H1) using a one-way ANOVA on trust in the recommendation. The results showed a significant main effect of explanations (F (2, 297) = 65.32, $p < 0.001$). Participants receiving either the consensual validation-based explanation (M = 5.32, SD = 1.12) or the expert validation-based explanation (M = 5.47, SD = 1.03) reported significantly higher trust than those receiving no explanation (M = 3.62, SD = 1.59, $p < 0.0001$). There was no significant difference in trust between the two explanation groups (p = 0.67).

We further investigated whether perceived interpretability and transparency mediate the effect of post hoc explanations on trust (H1a) using a parallel mediation analysis (Hayes, 2017, Model 4) with 5,000 bootstrapped samples. The results showed a significant indirect effect of post hoc explanations on trust in recommendations through perceived interpretability (b = 0.35, se = 0.04, 95% *CIs* [0.26, 0.44]) and perceived transparency (b = 0.17, se = 0.04, 95% *CIs* [0.10, 0.25]) (see Fig. 2).

5.3. Discussion

These results indicate that post hoc explanations significantly increase consumer trust in conversational recommendations (H1). Furthermore, this effect is mediated by perceptions of transparency and interpretability (H1a and H1b).

6. Study 2

Study 2 had three objectives: (1) to examine the interactive effects of explanation types and decision-making domains on consumer responses, (2) to investigate the three-way interaction between decision-making style, domain, and explanation type, and (3) to test the robustness of the findings by using smartwatches, in contrast to digital cameras in Study 1.

6.1. Method

Design and sample. We used a 2 (explanation type: user consensus-based vs. expert -based) × 2 (decision domain: hedonic vs. utilitarian) between-subjects design with a sample of 400 participants (58.0% female; M = 28.79 years, SD = 6.86) recruited from Credamo in exchange for 3 CNY.

Procedure and measures. Participants were asked to imagine purchasing a mid-range smartwatch using a conversational recommendation assistant. The assistant prompted participants to prioritize product attributes, offering utilitarian options (e.g., battery life, health monitoring) and hedonic options (e.g., appearance design, brand image), as identified in prior research (Voss et al., 2003). Participants were then assigned to either an expert-based explanation or a consensus-based explanation, corresponding to their decision context. Then, they rated perceived transparency (α = 0.86) and trust in the recommendation (α = 0.89), decision-making style (e.g., "I make decisions based on intuition," α = 0.86; Hernandez-Bocanegra and Ziegler, 2023). The full stimuli materials are available in Web Appendix D.

6.2. Results

To test whether the decision-making domain moderates the impact of explanations on consumer responses (H2), we conducted a 2 (consensual-based vs. expert-based explanation) × 2 (utilitarian vs. hedonic decision-making) MANOVA on perceived transparency and trust. The results showed significant interactions for both perceived transparency (F (1, 396) = 5.41, p = 0.021, η_p^2 = 0.01; see Fig. 3A) and trust (F (1, 396) = 4.34, p = 0.038, η_p^2 = 0.01; see Fig. 3B). Planned contrasts showed that consumers making hedonic decisions trusted consensus-based explanations more ($M_{\text{consensula-based}}$ = 5.56, $SD_{\text{consensula-based}}$ = 0.78 vs. $M_{\text{expert-based}}$ = 5.23, $SD_{\text{expert-based}}$ = 1.17, t (196) = 2.31, p = 0.003, d = 0.33). However, utilitarian consumers showed no significant trust differences between explanation types (t (200) = −0.65, p = 0.52). No significant main effects were found ($ps > 0.20$).

Moderated mediation analysis. To examine whether perceived transparency mediates the relationship between explanation type and trust, moderated by decision-making domain (H2), we conducted a PROCESS Model 8 analysis (Hayes, 2017) with 5000 bootstrapped samples: X = post hoc explanation type (0 = consensual-based, 1 = expert-based), M = perceived transparency, Y = trust, and W = decision-making domain (0 = rational, 1 = intuitive). The results indicated that consensus-based explanations significantly enhanced perceived transparency compared to expert-based explanations (b = −0.37, p = 0.007). Importantly, the interaction between explanation type and decision-making domain affected trust through perceived transparency (indirect effect = 0.30, se = 0.13; 95% *CIs* [0.05, 0.59]). The interaction effect for perceived transparency was significant (a_{inter} = 0.44, se = 0.19, t (396) = 2.33, p = 0.021). Path coefficients are presented in Fig. 3C.

Moderation by consumer decision-making style. We further tested whether decision-making style moderates the effect of explanation type and decision-making domain on trust, using PROCESS Model 12 (X = explanation type, M = perceived transparency, W = decision-making

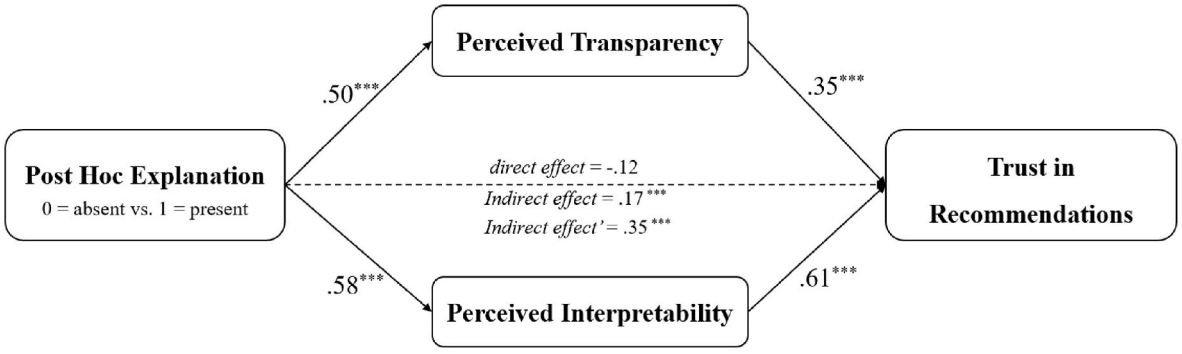
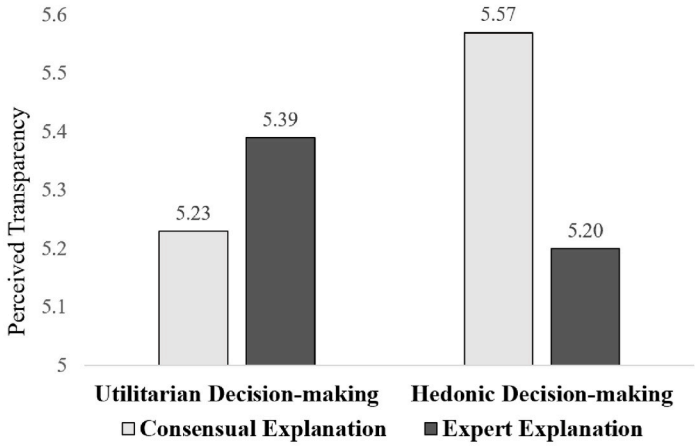
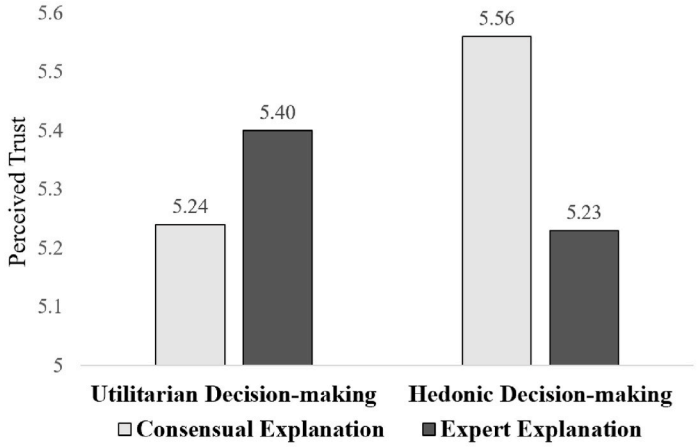


Fig. 2. Perceived interpretability and transparency mediate the effect of explanation on trust. ***$p < 0.001$. Coefficients are standardized.

A) Perceived transparency



B) Perceived trust



C) Mediation analysis

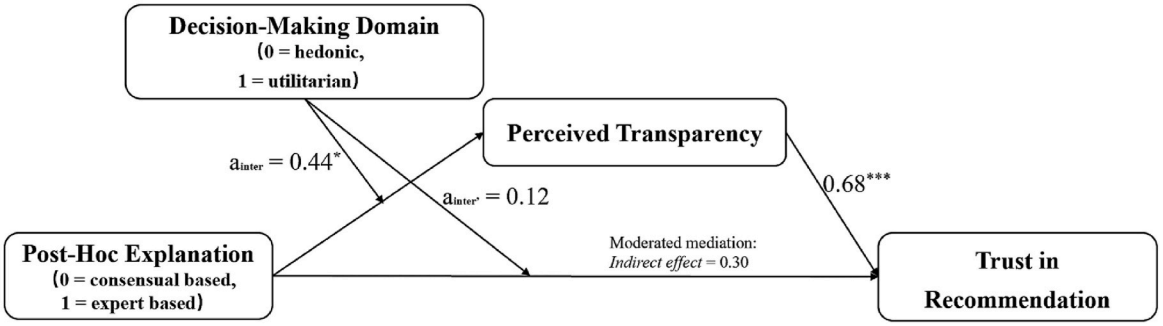


Fig. 3. Interactive effects of explanation type and decision-making domain on perceived transparency (panel A) and trust (panel B) in the recommendation in the mediation analysis (panel B). $^{*}p < 0.05$, $^{***}p < 0.001$.

domain, Z = decision-making style, and Y = trust, 5,000 bootstrapping iterations).

Results showed that the three-way interaction on trust was fully mediated by perceived transparency (indirect effect = 0.66, *se* = 0.26; 95% *CIs* [0.14, 1.16]). The three-way interaction effect was significant for perceived transparency ($b = 0.95$, $se = 0.37$, $t\ (392) = 2.58$, $p = 0.01$), which in turn influenced trust ($b = 0.70$, $se = 0.04$, $t\ (392) = 16.30$, $p < 0.001$). The interaction between explanation type and decision-making domain varied based on decision-making style. For intuitive consumers, the interaction effect was significant ($b = 1.00$, $p = 0.001$); however, it was not significant for rational consumers ($b = 0.06$, $p = 0.81$), indicating that decision style moderates the effect of decision domain on the relationship between explanation type and perceived transparency.

Participants were categorized as intuitive or rational decision-makers based on a median split of decision-making scores. Those scoring above the median were classified as intuitive, while those below the median were classified as rational. For intuitive decision-makers, there was a significant interaction between explanation type and decision-making domain ($F\ (1, 196) = 15.01$, $p < 0.001$, $\eta_p^2 = 0.07$; see Fig. 4A). Planned contrasts showed that consumers making hedonic decisions perceived consensus-based explanations as more transparent ($M_{\text{consensula-based}} = 5.49$, $SD_{\text{consensula-based}} = 0.69$ vs. $M_{\text{expert-based}} = 4.84$, $SD_{\text{expert-based}} = 0.99$, $t\ (105) = 3.97$, $p < 0.001$). Conversely, consumers making utilitarian decisions found expert-based explanations to be more transparent ($M_{\text{consensula-based}} = 4.83$, $SD_{\text{consensula-based}} = 0.82$ vs. $M_{\text{expert-based}} = 5.20$, $SD_{\text{expert-based}} = 1.10$, $t\ (91) = -1.74$, $p < 0.10$).

For rational decision-makers, this interaction was not significant ($F\ (1, 196) = 0.06$, $p = 0.81$; see Fig. 4B). Rational consumers consistently preferred consensus-based explanations, regardless of decision domain. All path coefficients are reported in Fig. 4C.

To further validate these findings, we conducted an additional study, in which participants selected their decision-making domain and preferred explanation type (see Web Appendix B).

6.3. Discussion

Study 2 demonstrated that the effectiveness of post hoc explanations depends on the decision-making domain (H2). User consensus-based explanations were more effective than expert-based explanations in hedonic decisions (H2a), while in utilitarian contexts, explanation type had no significant impact, rejecting H2b. Decision-making style moderated these effects (H3). Intuitive decision-makers favored consensus-based explanations in hedonic contexts and expert-based explanations in utilitarian contexts, whereas rational decision-makers consistently preferred consensus-based explanations, regardless of decision domains.

7. Study 3

Study 3 examined whether the user interface moderates the matching effect between the explanation type and the decision-making domain. Participants completed a survey using either a text-based or voice-based interface of a conversational recommendation agent.

7.1. Method

Design and sample. We employed a 2 (explanation type: consensual-based vs. expert-based) × 2 (decision domain: hedonic vs. utilitarian) × 2 (interface modality: text-based vs. voice-based) between-subjects design. A total of 729 participants were recruited through Credamo (62.9% female, $M_{\text{age}} = 28.26$, $SD = 7.40$).

Procedure and measures. Participants were randomly assigned to either a voice-based or text-based interface. The text-based condition followed the format used in previous studies, while the voice-based condition simulated audible recommendations using a smart voice assistant. Participants were introduced to real-world examples of smart voice assistants, such as Amazon's Echo and Alibaba's Tmall Genie. We utilized a text-to-speech demonstration available at https://saas.xfyun.cn/speech?. For consistency, all voice-based conditions featured a standard adult female voice, and the content and sequence of voice-based conversations mirrored those in the text-based condition. Participants then reported their intention to follow the AI recommendation (e.g., "I am likely to follow the recommendation obtained from the AI assistant," α = 0.82), adapted from Flavián et al. (2023).

7.2. Results

We conducted a 2 (explanation type: consensual-based vs. expert-based) × 2 (decision domain: hedonic vs. utilitarian) × 2 (interface modality: text-based vs. voice-based) ANOVA on the intention to follow AI recommendations. The results indicated a significant three-way interaction ($F\ (1, 721) = 10.00$, $p = 0.002$, $\eta_p^2 = 0.014$). To further analyze this effect, we conducted separate ANOVAs for text-based and voice-based interfaces.

The results confirmed previous findings. A significant interaction effect was noted under the text-based interface ($F\ (1, 369) = 5.11$, $p = 0.024$, $\eta_p^2 = 0.014$; see Fig. 5A). As we predicted, the voice-based interface showed the opposite pattern ($F\ (1, 352) = 4.98$, $p = 0.026$, $\eta_p^2 = 0.014$; see Fig. 5B). For utilitarian decisions, consumers were more likely to follow AI recommendations when given a consensual-based explanation ($M_{\text{consensual-based}} = 5.74$, $SD_{\text{consensual-based}} = 0.64$ vs. $M_{\text{expert-based}} = 5.54$, $SD_{\text{expert-based}} = 0.70$, $t\ (175) = -2.02$, $p = 0.045$, $d = -0.30$). For hedonic decisions, no significant difference was found between the two explanation types ($M_{\text{consensual-based}} = 5.50$, $SD_{\text{consensual-based}} = 0.83$ vs. $M_{\text{expert-based}} = 5.63$, $SD_{\text{expert-based}} = 0.70$, $t\ (177) = 1.21$, $p = 0.23$, $d = 0.18$). The user interface significantly affected consumer responses, with a greater intention to follow AI recommendations in the voice-based interface ($F\ (1, 721) = 16.29$, $p < 0.001$).

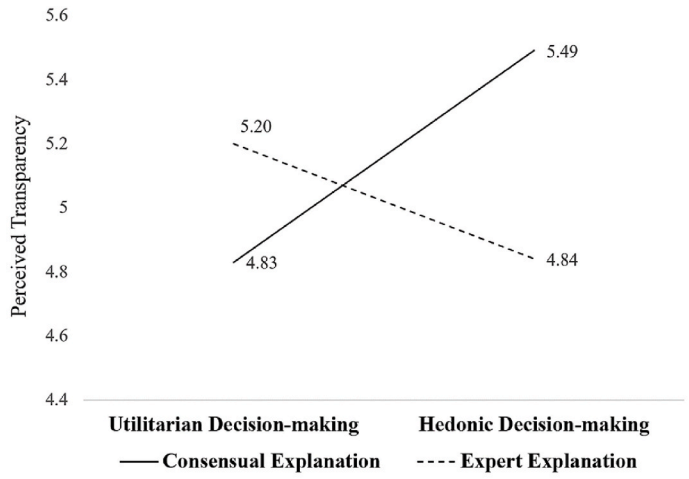
7.3. Discussion

Study 3 findings emphasize the moderating role of the user interface in the relationship between explanation type and decision-making domain (H4). Specifically, the voice-based interface increased consumers' intention to follow AI recommendations compared to the text-based interface. This aligns with previous research on the effectiveness of voice-based recommendations in enhancing consumer experiences (Akdim and Casaló, 2023; Flavián et al., 2023). Additionally, strong effects on purchase intention (see Web Appendix C) further validate our findings. Notably, while Studies 1 and 2 focused on trust as a key attitudinal indicator, we omitted trust in Study 3 to instead examine intention to follow recommendation—a more direct behavioral outcome relevant for assessing interface effects. This shift allowed us to capture a new dimension of positive consumer response, highlighting how interface modality influences consumer behavior in AI-driven recommendations.

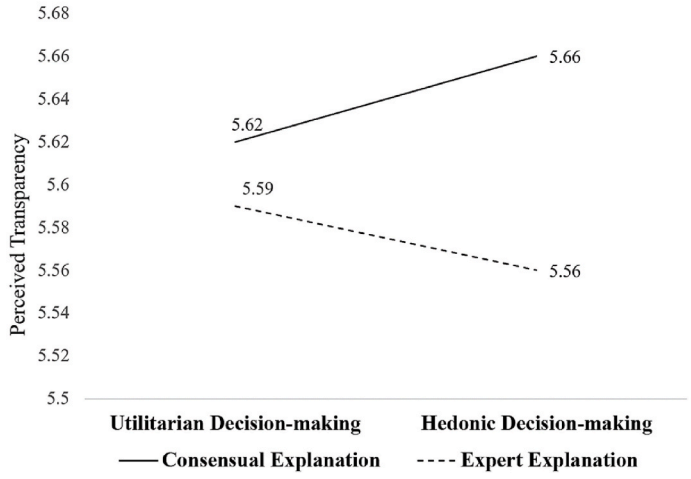
8. General discussion

As conversational recommendation agents become increasingly integrated into e-commerce platforms, understanding how different post hoc explanation types influence consumer behavior is critical for both academics and practitioners. This research investigates how aligning explanation types (consensual validation-based vs. expert validation-based) with decision-making domains (hedonic vs. utilitarian) impacts consumer responses in conversational recommendation settings.

Across three studies, the findings demonstrate that post hoc explanations enhance interpretability and transparency, thereby increasing consumer trust in conversational recommendations. Consumers generally favor consensus-based explanations in hedonic contexts, whereas utilitarian decision-makers exhibit no strong preference between



B) Consumers with rational decision-making style



C) Moderated mediation analysis

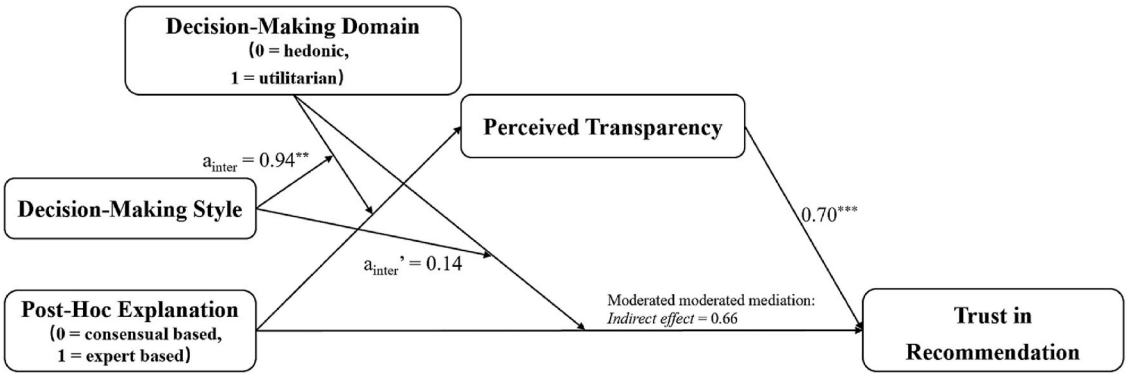


Fig. 4. Interactive effect of post hoc explanation type, decision-making domain and decision-making style on perceived transparency (Panels A and B) in the mediation analysis (Panel C). $^{**}p < 0.01$, $^{***}p < 0.001$.

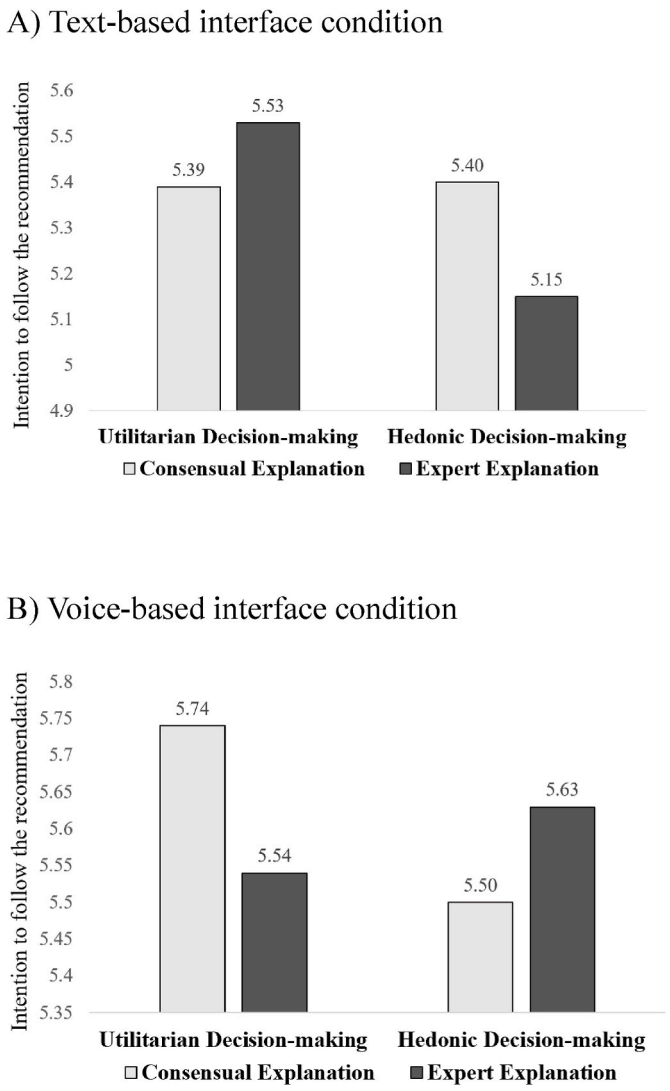


Fig. 5. Interactive effect of post hoc explanation type, decision-making domain and user interface on the intention to follow the AI recommendation in Study 3.

explanation types. Decision-making style further moderates this effect—intuitive consumers prefer consensus-based explanations in hedonic decisions and expert-based explanations in utilitarian decisions, whereas rational consumers consistently favor consensus-based explanations across both domains. Notably, in voice-based interfaces, the preference for consensus-based explanations shifts from hedonic to utilitarian decisions, underscoring the role of interface modality in shaping consumer preferences.

8.1. Theoretical contributions

This study advances research in retailing and consumer services by offering new insights into how explainability in AI-driven recommendations influences consumer decision-making.

First, this research expands the literature on post hoc explainability in conversational recommender systems. While prior work has primarily examined static explainable recommendation systems (Guo et al., 2023a,b), these findings underscore the unique role of conversational recommendation agents in fostering consumer trust through real-time, dynamic explanations. This study bridges a critical gap by highlighting how conversational interactions shape consumer perceptions and decision-making in digital retail environments.

Second, the study demonstrates that aligning the explanation type with the decision-making domain is essential for enhancing consumer responses. Previous research has established that decision contexts influence consumer information processing (Chen et al., 2024); however, these findings extend this understanding by showing that matching explanation type to decision domain improves consumer trust and engagement in AI-driven retail environments. This insight has important implications for e-commerce platforms seeking to optimize recommendation systems for different product categories.

Third, this research contributes to the growing literature on consumer decision-making styles in AI-driven retail environments. Recent studies emphasize the importance of tailoring recommendation systems to individual user preferences (EI Aouifi et al., 2023). These findings reveal that intuitive consumers respond more favorably to consensus-based explanations in hedonic contexts and expert-based explanations in utilitarian contexts, whereas rational consumers consistently prefer consensus-based explanations. These findings highlight the need for personalized AI-driven recommendation strategies, aligning with broader trends in retail personalization and adaptive recommendation models.

Finally, this study advances the field of multimodal recommendation systems by examining the impact of interface modality on consumer responses. Prior studies on explainability in conversational recommendations have primarily focused on text-based systems (Gao et al., 2021). The results indicate that in voice-based interfaces, consensus-based explanations become more effective for utilitarian decisions, reversing the trend observed in text-based interactions. This highlights the critical role of interface design in shaping consumer trust and recommendation adoption, offering new insights for retailers integrating voice-assisted AI into digital shopping experiences.

8.2. Practical implications

This research provides several actionable insights for e-commerce retailers, AI developers, and UX designers aiming to optimize conversational recommendation systems for enhanced consumer engagement and trust.

First, tailoring post hoc explanations to the decision-making domain can enhance consumer trust and engagement. Retailers implementing conversational recommendation agents should recognize that hedonic consumers—who prioritize experiential and sensory attributes—prefer consensus-based explanations, whereas utilitarian consumers exhibit no strong preference between explanation types. Providing both consensus-based and expert-based explanations allows consumers to select the approach that best aligns with their decision-making style, ultimately improving the effectiveness of AI-driven recommendations. Platforms like Amazon and Alibaba, which sell both hedonic and utilitarian products, can benefit from context-sensitive explainability strategies to increase consumer confidence and conversion rates.

Second, AI-driven recommendation systems should incorporate consumer decision-making styles into personalized shopping experiences. These findings suggest that intuitive consumers prefer explanations that align with their affect-driven decision-making, such as consensus-based explanations in hedonic contexts and expert-based explanations in utilitarian contexts. Rational consumers, however, favor consensus-based explanations regardless of the decision domain. By leveraging consumer behavioral data, retailers can dynamically tailor recommendation explanations based on individual decision styles, leading to greater consumer satisfaction, engagement, and long-term loyalty.

Third, businesses leveraging voice-based AI assistants must consider interface modality when designing recommendation explanations. In text-based interfaces, consumers prefer consensus-based explanations in hedonic contexts; however, in voice-based interactions, this preference shifts to utilitarian decisions. Given the increasing adoption of voice-enabled shopping (e.g., Amazon Alexa, Apple Siri, Google Assistant), e-commerce platforms should strategically adjust recommendation

explanations to align with the interface modality. Retailers incorporating voice-based conversational AI into online and omnichannel shopping experiences can enhance consumer trust and drive purchasing behavior by leveraging interface-specific recommendation strategies.

Finally, offering transparent and interpretable explanations in AI-driven recommendations strengthens consumer trust in digital retail environments. By enhancing perceived interpretability and transparency, retailers can mitigate skepticism toward AI-driven recommendations and improve consumer attitudes toward automated decision-making systems. Industries with complex decision-making processes, such as electronics, healthcare, and financial services, may particularly benefit from expert-based explanations that validate technical specifications and data-driven insights.

8.3. Limitations and future research

First, this study focused exclusively on Chinese consumers. Given that collectivist cultures emphasize social validation (Cialdini and Goldstein, 2004), consensus-based explanations may be more persuasive in this context. Conversely, individualist cultures (e.g., the United States) tend to prioritize expert opinions and autonomy (Hofstede, 2011). Future research should explore cross-cultural variations in consumer responses to AI-driven recommendation explanations.

Second, expert-based explanations may be more effective in specialized decision contexts. The findings suggest that expert-based explanations were not significantly more persuasive than consensus-based explanations in e-commerce settings. However, in domains requiring specialized knowledge, such as medicine, finance, and legal services, expert-based explanations may be more critical for consumer trust and decision-making (Bonaccio and Dalal, 2006). Future research should investigate context-dependent preferences for expert validation across industries.

Third, this study focused on text-based and voice-based interfaces, excluding emerging interaction modalities. With the rise of augmented reality (AR), virtual reality (VR), and gesture-based interfaces, future studies should examine how these immersive experiences affect consumer responses to explainable AI recommendations (Pjesivac et al., 2023).

Fourth, this research examined only consensus-based and expert-based explanations. Other explanation types, such as empirical validation (e.g., “This product is recommended based on usage data”) or internal validation (e.g., “This recommendation aligns with your past purchases”), may be more effective for high-involvement or data-driven purchases (Jaccard and Jacoby, 2009). Future research should explore how different explanation types interact with consumer traits, decision contexts, and shopping motivations.

8.4. Conclusion

As conversational recommendation agents continue to transform digital retail, understanding how consumers respond to explanation types is essential for optimizing AI-driven recommendation strategies. This study provides theoretical insights and practical implications for retailers and developers, demonstrating the importance of aligning explanation type with decision domain, consumer decision-making style, and user interface modality. By integrating these insights, e-commerce platforms can enhance consumer trust, engagement, and satisfaction, ultimately driving higher conversion rates and long-term consumer loyalty.

CRediT authorship contribution statement

Qian Qian Chen: Writing – original draft, Funding acquisition, Formal analysis, Conceptualization. **Li Min Lin:** Validation, Methodology, Investigation. **Youjae Yi:** Writing – review & editing, Supervision, Project administration.

Declaration of competing interest

The authors declare that they have no conflict of interest.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jretconser.2025.104281.

Data availability

Data will be made available on request.

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